

DETAILED PROJECT REPORT

150 kW DC Fast-Charging Station, Gadag

Technical design, project cost, market study and ten-year financial projections for term-loan appraisal

Borrower / promoter	Mr. Manjunath
Project	1 × 120 kW + 1 × 30 kW dual-gun DC fast chargers (Exicom), canopy, 200 kVA HT substation
Location	Hubli–Gadag Road (NH-67), Binkadakatti, Gadag taluk, Karnataka · 15.42335° N, 75.57571° E
Project cost / funding need	₹54.61 lakh project cost · ₹70.61 lakh with operating reserve
Proposed term loan	₹38.20 lakh · 10 years incl. 24-month moratorium
Prepared by	Sonar.EV

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How to read this report

Each figure carries a tag showing where it comes from, so the lender can see what is evidence and what is estimate.

SOURCED

Published source, linked in Appendix C.

QUOTED

Vendor quotation: The Lynx, dated 06-10-2026.

CLIENT

Supplied by the client (charger and canopy prices, GST rate, site markup).

ASSUMED

Sonar.EV assumption. It must be replaced by a measured or quoted figure before final sanction.

Disclaimer

This report was prepared from the documents and data listed above and public sources available on 8 October 2026. Demand, tariff and traffic inputs are estimates, not guarantees, and no traffic survey has yet been carried out. Projections are for appraisal and are not a forecast or assurance of results. Blank fields marked _____ are to be completed by the borrower before submission.

1. Executive summary

Mr. Manjunath proposes a public DC fast-charging station on the Hubli–Gadag Road (NH-67), about 5.8 km from Gadag town. It will have one 120 kW and one 30 kW dual-gun Exicom charger (four vehicles at once), a canopy and a new 200 kVA, 11 kV substation. The project costs **₹54.61 lakh** including 5% GST and a statutory-fee allowance, and the promoter seeks a **₹38.20 lakh term loan (70%)**.

₹54.6 L

Project cost incl. GST

₹38.2 L

Term loan sought (70%)

20%

Project IRR, 10 yrs (base)

3.1×Avg DSCR, repayment
years (base)

Key points for the lender

- **Fixed, quoted costs.** 95% of project cost is backed by a vendor quotation or client-confirmed prices. A statutory-fee allowance of ₹3 lakh covers items the quotation lists at actuals.
- **Growing market.** Karnataka registered 24,201 electric cars in FY 2025-26, about 11% of India's total, nearly double the 12,200 of the year before. Only two charging points are listed in Gadag, and none is a named DC fast charger.
- **Break-even is low.** The station covers its running costs at about 7 sessions a day (159 kWh, 4.4% use of 150 kW). The base case reaches that level in year 3.
- **Early years are loss-making.** EBITDA is negative in years 1 and 2 in the base case, so DSCR is below 1.0 until year 4. The structure therefore has a 24-month principal moratorium, step-up repayments and a ₹16 lakh promoter reserve to carry interest through the ramp-up.
- **Largest unknown: traffic.** No published count exists for this stretch. The model assumes 12,000 vehicles a day. A 7-day classified count (template in Appendix B) should be done before sanction.

Downside. In the conservative case (8,000 vehicles a day, fewer EVs, fewer stops) the project does not repay from operations within 10 years. The lender should treat traffic and EV adoption as the main credit risk.

2. Borrower profile and bank-readiness checklist

Fields in grey are for the borrower to complete. Nothing in this section has been assumed.

Borrower	Mr. Manjunath
Constitution	Individual / proprietorship / firm: _____
PAN · Udyam registration no.	_____ · _____
GSTIN	_____
Residential / business address	_____
Experience and occupation	_____
Net worth (as on _____)	₹ _____
Existing bank borrowing	₹ _____ with _____
CIBIL score of promoter	_____
Site owned / leased	Owned / leased: _____; survey no. _____; extent _____; land-use status _____

Documents the bank will typically ask for

Document	Status
☐ KYC of promoter and co-applicants; PAN; Aadhaar	To obtain
☐ Land title, RTC/khata, encumbrance certificate; land-use conversion (if required)	To obtain
☐ Udyam registration; GST registration	To obtain
☐ ITR and computation of income, last 3 years; bank statements, last 12 months	To obtain
☐ Statement of assets and liabilities; net worth certificate	To obtain
☐ Vendor quotations: The Lynx (06-10-2026); Exicom charger quote; canopy quote	Lynx on file; others to obtain
☐ Distribution-company feasibility / sanction; CEIG approval application	To obtain
☐ NHAI/PWD permission for road access; local authority NOC	To obtain
☐ Traffic survey report (7-day, classified); site photographs	To obtain
☐ Insurance quotation for the station (fire, equipment, third-party)	To obtain
☐ Charge-point operator agreement and payment-gateway agreement	To obtain
☐ This DPR with financial model	To obtain

3. Project description and location

The project is a roadside public charging station for electric cars, light commercial vehicles and buses on NH-67, the trunk road that links Dharwad, Hubballi, Gadag, Koppal, Hosapete and Ballari (en.wikipedia.org). The plot is in Binkadakatti village, Gadag taluk, on the north side of the road, with 60 ft of frontage **CLIENT**.

Item	Detail
Coordinates	15.42335° N, 75.57571° E (from the shared map link)
Road	Hubli–Gadag Road, classified trunk (OpenStreetMap)
Distance to nearby towns (straight line, approx.)	Gadag 5.8 km · Hubballi 48.9 km · Dharwad 61 km · Koppal 63 km · Hosapete 89 km
Site frontage	60 ft CLIENT ; depth to be measured
Surroundings	Open agricultural land on both sides; small building and residential layout plots to the east



Figure 1. Regional view (~3.5 km across)

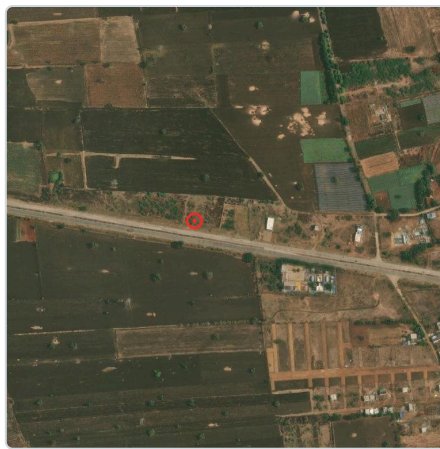


Figure 2. Corridor view (~880 m)

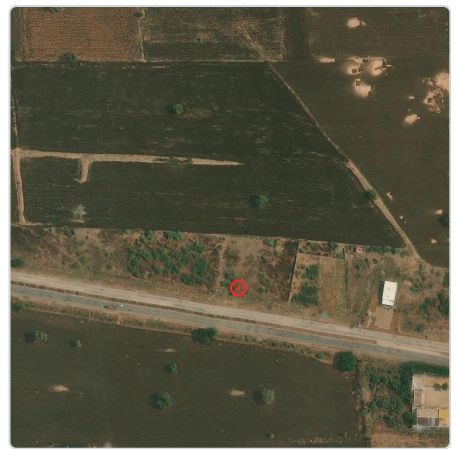


Figure 3. Site close-up (~440 m). Red ring = plot.

Imagery: Esri World Imagery (© Esri, Maxar, Earthstar Geographics and the GIS user community), retrieved 8 Oct 2026; capture dates vary.

4. Site survey and layout



Figure 4. Client's site markup: 60 ft frontage on Google Maps (scale 50 ft / 20 m)

Survey findings

- Direct frontage on a four-lane trunk road; entry and exit from the road side.
- Flat, open land with light vegetation.
- HT supply must be extended to the plot: 44 m of 11 kV cable and a pole-mounted GOS are in the quotation **QUOTED**.
- Hard-soil trenching allowed for 26 m³ and soft-soil trenching 33 m³ **QUOTED**.
- Fire kit, bollards, car stoppers, fencing and signage are included.

Open items: plot depth and boundary sketch; nearest HT pole and distance; soil test; confirmation that the existing quotation quantities match the final plot; NHAI/PWD road-entry permission.

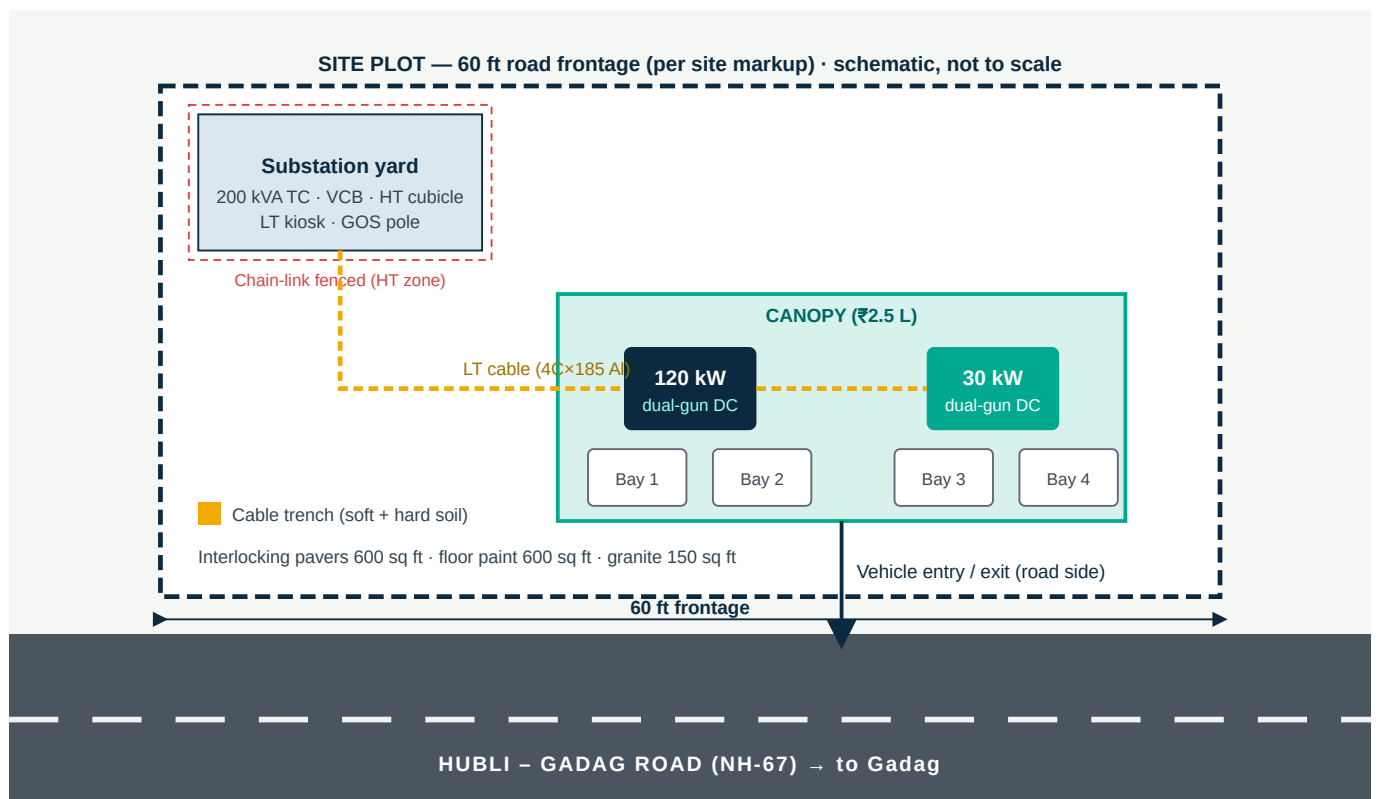


Figure 5. Schematic site plan, not to scale.

5. Technical design

Item	Specification	Basis
Chargers	Exicom DC fast chargers, dual-gun: 1 × 120 kW and 1 × 30 kW (150 kW total, 4 guns)	CLIENT
Charger make	Exicom. Exicom states that its Harmony DC range has efficiency above 95% and modular power from 60 kW to 400 kW (exicom.in). The model datasheet (connector, OCPP version, IP rating) is to be attached once quoted.	SOURCED
HT supply	11 kV line, GOS 400 A with lightning arrester, HT metering cubicle, 11 kV 630 A 25 kA VCB	QUOTED
Transformer	200 kVA, 11/0.433 kV, ONAN, oil-filled, off-load tap changer, to IS 1180	QUOTED
LT distribution	400 A 415 V kiosk; 250 A 4P MCCB for 120 kW, 63 A 4P MCCB for 30 kW; 3.5C × 185 mm ² Al cable to chargers	QUOTED
Earthing	12 copper-bonded pipe electrodes (HT yard), 3 chemical electrodes (chargers), GI flat 50×6 mm, copper plate 600×600 mm for transformer neutral	QUOTED
Civil	RCC M20 charger foundation 4700×1000×400 mm; equipment foundations; 600 sq ft interlocking pavers; 600 sq ft painted bays; 150 sq ft granite	QUOTED
Canopy	Steel canopy over charging bays	CLIENT
Safety	Fire extinguishers (ABC), fire buckets, danger boards, shock-treatment chart, cable markers, bollards, car stoppers, chain-link fence	QUOTED

Load check. The two chargers draw up to 150 kW of DC output. At typical conversion efficiency of 95% (Exicom's published figure for its range) that is about 158 kVA at unity power factor, which sits within the 200 kVA transformer with headroom for auxiliaries. The vendor must confirm the exact input rating of the chosen models.

Single-line power flow: 11 kV supply to the two chargers

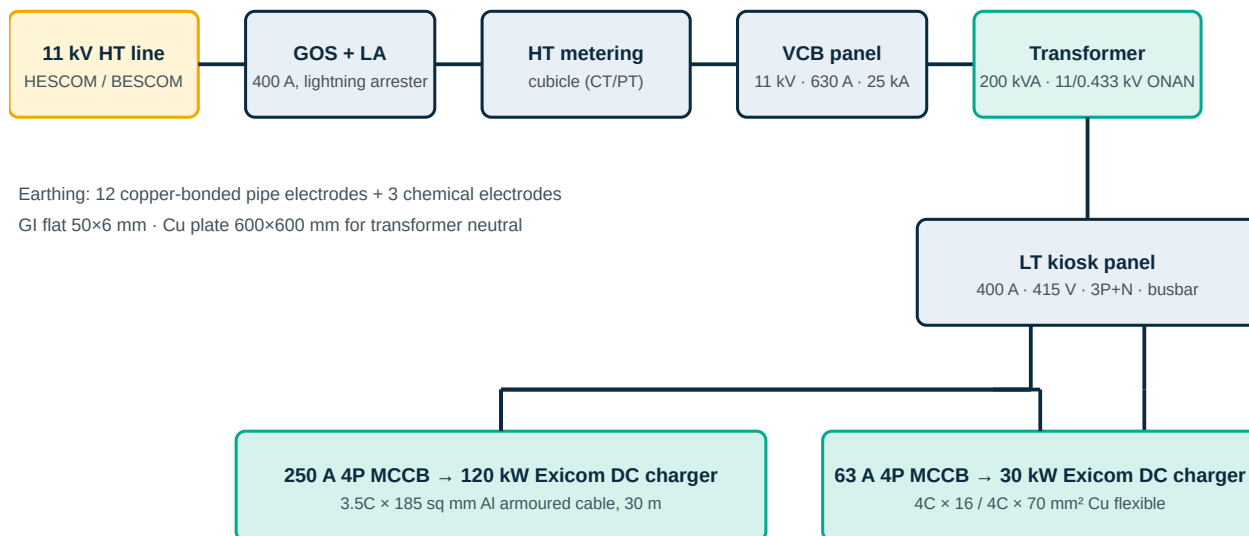


Figure 6. Single-line power flow.

The quotation names BESCOM for sanction. Gadag district is normally served by HESCOM; the distribution company must be confirmed before the application is filed.

6. Project cost and bill of quantities

Component	Amount (₹)	Basis
Electrical and civil infrastructure (59 items; see Appendix A)	32,04,798	QUOTED
Canopy	2,50,000	CLIENT
30 kW Exicom dual-gun DC charger	4,60,000	CLIENT
120 kW Exicom dual-gun DC charger	10,00,000	CLIENT
Project cost before GST	49,14,798	
GST @ 5%	2,45,739.90	CLIENT
Cost including GST	51,60,537.90	
Statutory deposits, inspection fees and contingency (allowance)	3,00,000	ASSUMED
Total project cost	54,60,538	

The quotation lists security deposit, CEIG, CT/PT testing and TAQC/MRT fees "at actuals" with no amount. The allowance is a placeholder and must be replaced with the distribution company's demand note. GST: the quotation states 18%; the client has confirmed 5%, which must be matched to the final invoices. If the borrower is GST-registered, input credit rules should be confirmed with a tax adviser.

Quoted infrastructure by category (₹32,04,798)

Category	Amount (₹)	Share
HT equipment (GOS, cubicle, VCB, transformer)	11,72,000	36.6%
Civil, site and finishing	6,86,426	21.4%
Cables, terminations and pipes	3,18,384	9.9%
Statutory and approvals	3,05,000	9.5%
Earthing	2,87,738	9.0%
LT panels and installation	2,50,000	7.8%
Project management and logistics	1,60,000	5.0%
Safety and fire	25,250	0.8%
Total	32,04,798	100%

Cost per kW. ₹0.33 lakh per kW of installed charging power (before GST), which includes the HT substation. Sites that share an existing transformer have a lower figure; this one carries the full grid-connection cost.

7. Means of finance

Means of finance — total ₹70.61 lakh



Source	₹	Share	Terms
Term loan	38,20,000	54%	10 years incl. 24-month moratorium on principal; interest serviced monthly; step-up repayment (5%, 7.5%, 10%, 12.5%, 12.5%, 15%, 17.5%, 20% over years 3–10)
Promoter equity (margin on project cost)	16,40,538	23%	30% of project cost, paid upfront
Promoter operating-deficit reserve	16,00,000	23%	Covers interest and running costs in years 1–3 (base case cumulative shortfall ₹15.42 lakh); to be held in an escrow or fixed deposit
Total funds	70,60,538	100%	

Reference terms (published)

- An MSME loan offered through a Tata Power Solar and Bank of India tie-up for EV charging stations carried interest from 9.35%, 15% margin and tenor up to 10 years, as announced in July 2024 (www.pv-magazine-india.com). Terms may have changed.
- Public-sector banks have proposed priority-sector treatment for charging infrastructure, up to ₹2.5 crore per borrower (August 2026, proposal only; www.alcircle.com). It is not yet policy.

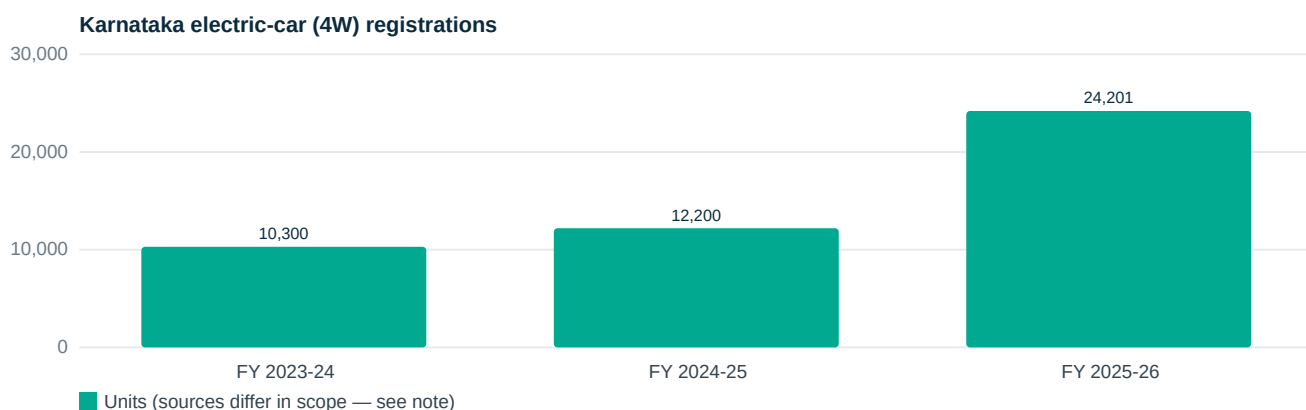
Interest assumption. **ASSUMED** The model uses 10.5% a year, above the 2024 reference, to leave room for the lender's actual pricing. Please replace it with the sanctioned rate.

Possible central subsidy (not in the base case). PM E-DRIVE earmarks ₹2,000 crore for about 72,300 public chargers, including 22,100 fast chargers for electric four-wheelers on 50 national-highway corridors. Reports say the scheme funds 80% of upstream infrastructure at public and highway locations, and the guidelines work through nominated state agencies (www.outlookbusiness.com). Whether a private plot qualifies must be checked against the official guidelines; no subsidy is counted here.

8. Market study

India and Karnataka

- India registered about 23 lakh electric vehicles in calendar 2025, 8% of all new vehicles; electric four-wheelers were 1.75 lakh units ([newsonair.gov.in](https://www.newsonair.gov.in)).
- Karnataka sold 24,201 electric cars in FY 2025-26, second in India with about 11% of the national total; Tata Motors (8,348) and JSW MG Motor (8,157) led (evreporter.com).
- Electric four-wheeler registrations in Karnataka rose from 10,300 in FY 2023-24 to 12,200 in FY 2024-25 (www.deccanherald.com). The same source puts Karnataka's overall EV share at 9.38% in FY 2025-26 against 7.49% nationally; Down To Earth reports 11%. Definitions differ.
- Karnataka's road-tax exemption for electric cars ended on 1 April 2026; new cars now pay lifetime tax of 5–10% by price band (as reported by vahanbazaar.in). This may slow private purchases, but it does not affect fleet or visiting vehicles from other states.



The FY 2023-24 and 2024-25 figures come from Deccan Herald and FY 2025-26 from EVreporter's Vahan analysis. The sources differ in scope, so read the growth as indicative.

Gadag and the corridor

- Gadag district had 10,64,570 people at the 2011 census; Gadag-Betageri city had about 1.73 lakh (www.citypopulation.de; en.wikipedia.org). Registered vehicle counts for the district (KA-26) were not found in public sources.
- A public directory lists two charging points in Gadag: an Ather station on Hubli Road and an IOCL station on Mulgund Road, with none described as DC fast (www.91trucks.com). On-site verification is needed.
- NH-67 is a freight and tourist corridor to Hampi. NHAI awarded a 144 km Hubballi–Hosapete section that passes through Gadag (www.itln.in).

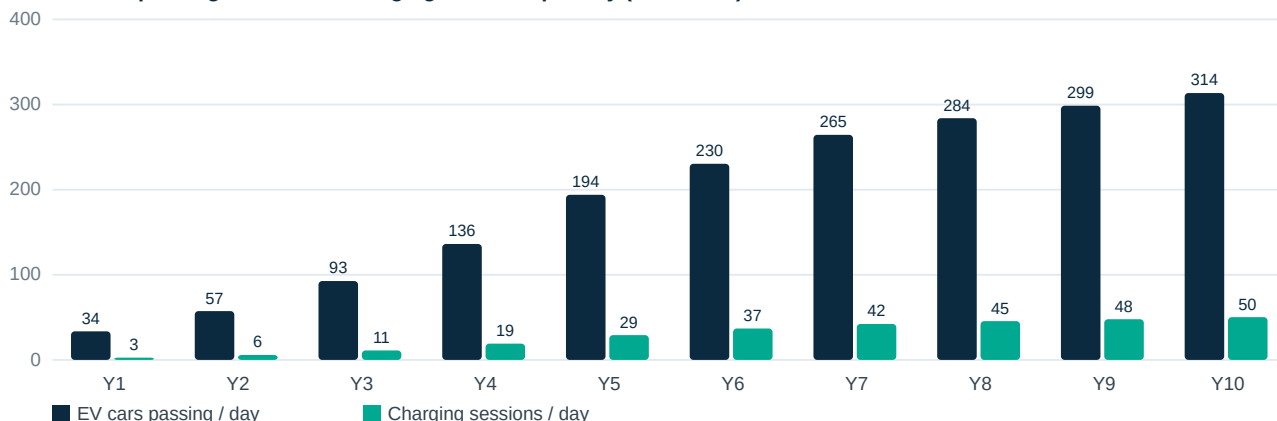
Tariff context

Karnataka Electricity Regulatory Commission's 2025 tariff order treats EV charging as a separate category (LT-6c) and notes it is cross-subsidised (www.energetica-india.net). The HT energy and demand charges that apply to an 11 kV EV station were not found; the model uses ₹7 per kWh all-in and must be verified with the distribution company.

9. Traffic and demand model; assumptions register

Passing EV cars = vehicles per day × 35% cars × electric share. Sessions = passing EVs × stop rate. Energy sold = sessions × 24 kWh. The model has no traffic input other than the assumed 12,000 vehicles a day, so the largest uncertainty is easy to test: every 1,000 vehicles per day adds or removes about ₹4.2 lakh of year-5 revenue.

EV cars passing the site vs. charging sessions per day (base case)



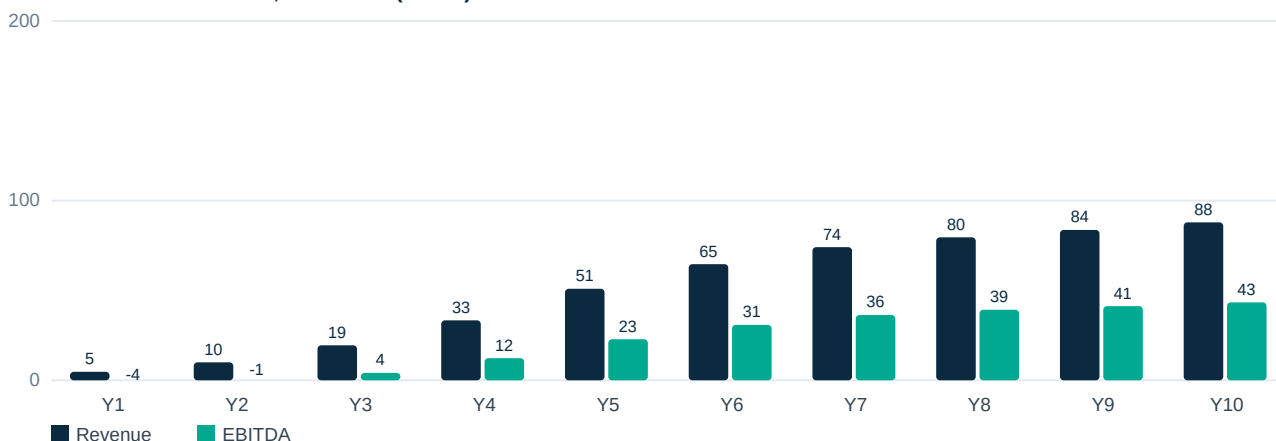
Input	Base value	Basis	Note
Vehicles per day at the site (year 1)	12,000, +5% a year to year 7, then +3%	ASSUMED	No traffic count found for this stretch. To be replaced by a 7-day classified count
Cars in the traffic mix	35%	ASSUMED	Typical national-highway mix; confirm in the count
Electric share of passing cars	0.8% in Y1 → 4.7% in Y7 → 5.1% in Y10	ASSUMED	Anchored on Karnataka e-car growth; fleet share lags sales share
Share of passing EVs that stop to charge	8% in Y1 → 16% from Y6	ASSUMED	Rises with awareness; no named DC fast charger nearby
Energy per session	24 kWh	ASSUMED	Typical highway top-up
Selling price	₹20 per kWh before GST	ASSUMED	Market-typical public DC price; owner sets
Power cost	₹7 per kWh all-in	ASSUMED	To be confirmed with distribution company
Running costs	₹7.2 L in Y1, +5% a year	ASSUMED	Two attendants, maintenance contract, insurance, connectivity; land owned
Payment and software fees	3% of revenue	ASSUMED	Gateway and charge-point management
Depreciation	10 years straight line on ₹49.15 L	ASSUMED	Simplified; tax depreciation will differ
Income tax	25%, with losses carried forward	ASSUMED	Simplified
Loan interest	10.5% a year	ASSUMED	See section 7
Capex, GST rate	As quoted / confirmed	QUOTED	See section 6

Every input marked ASSUMED is an estimate. A lender will want the traffic count, a distribution-company tariff letter and a signed charger quote before relying on the projections below.

10. Financial projections (base case)

Revenue reaches ₹51.0 lakh in year 5 and ₹87.9 lakh in year 10. EBITDA turns positive in year 3. Figures are in rupees, exclude GST on sales, and ignore working-capital swings (customers pay at the charger).

Revenue and EBITDA, base case (₹ lakh)



₹ (INR)	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10
Revenue	4,70,938	10,04,422	19,47,033	33,39,161	50,98,183	64,61,276	74,15,442	79,62,922	83,69,194	87,92,675
Power cost	-1,64,828	-3,51,548	-6,81,461	-11,68,706	-17,84,364	-22,61,447	-25,95,405	-27,87,023	-29,29,218	-30,77,436
Payment fees	-14,128	-30,133	-58,411	-1,00,175	-1,52,945	-1,93,838	-2,22,463	-2,38,888	-2,51,076	-2,63,780
Running costs	-7,20,000	-7,56,000	-7,93,800	-8,33,490	-8,75,165	-9,18,923	-9,64,869	-10,13,112	-10,63,768	-11,16,956
EBITDA	-4,28,019	-1,33,259	4,13,360	12,36,790	22,85,709	30,87,069	36,32,705	39,23,899	41,25,132	43,34,502
Depreciation	-4,91,480	-4,91,480	-4,91,480	-4,91,480	-4,91,480	-4,91,480	-4,91,480	-4,91,480	-4,91,480	-4,91,480
Interest	-4,01,100	-4,01,100	-4,01,100	-3,81,045	-3,50,962	-3,10,852	-2,60,715	-2,10,578	-1,50,412	-80,220
Profit before tax	-13,20,598	-10,25,838	-4,79,220	3,64,265	14,43,267	22,84,736	28,80,510	32,21,842	34,83,240	37,62,802
Tax @25%	0	0	0	0	0	-3,16,653	-7,20,128	-8,05,461	-8,70,810	-9,40,701
Profit after tax	-13,20,598	-10,25,838	-4,79,220	3,64,265	14,43,267	19,68,083	21,60,383	24,16,382	26,12,430	28,22,102

Operating metrics	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10
Vehicles per day	12,000	12,600	13,230	13,892	14,586	15,315	16,081	16,564	17,060	17,572
Sessions per day	2.7	5.7	11.1	19.1	29.1	36.9	42.3	45.5	47.8	50.2
kWh sold per day	65	138	267	457	698	885	1,016	1,091	1,146	1,204
Use of 150 kW*	2%	4%	7%	13%	19%	25%	28%	30%	32%	33%

*Energy sold ÷ (150 kW × 24 h). Indian public chargers typically run well below 20% in early years; the base case passes that level only in year 5.

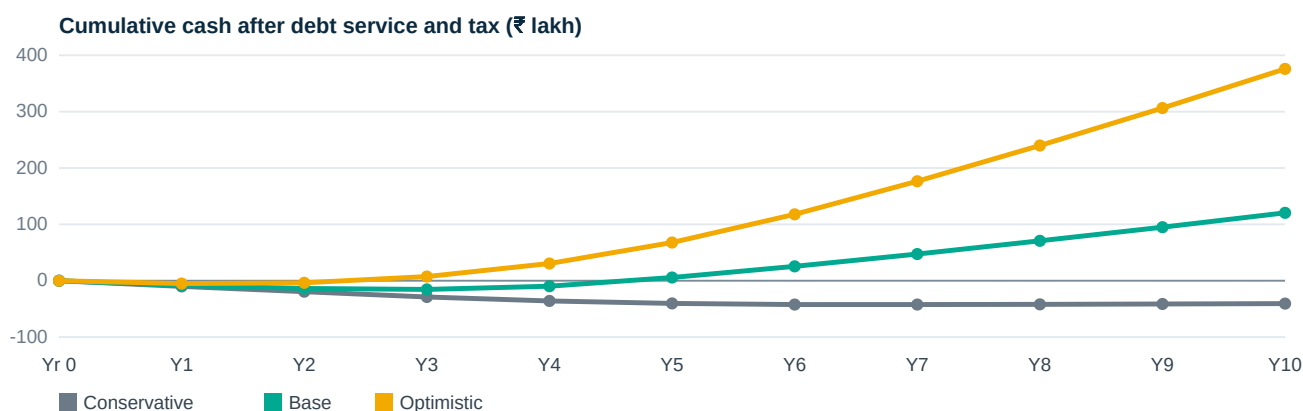
11. Debt service, DSCR and returns

Loan ₹38.20 lakh at 10.5%, 24-month principal moratorium, step-up repayment from year 3. The loan is repaid in full by year 10.

₹	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10
Opening balance	38,20,000	38,20,000	38,20,000	36,29,000	33,42,500	29,60,500	24,83,000	20,05,500	14,32,500	7,64,000
Principal repaid	0	0	1,91,000	2,86,500	3,82,000	4,77,500	4,77,500	5,73,000	6,68,500	7,64,000
Interest @10.5%	4,01,100	4,01,100	4,01,100	3,81,045	3,50,962	3,10,852	2,60,715	2,10,578	1,50,412	80,220
Total debt service	4,01,100	4,01,100	5,92,100	6,67,545	7,32,962	7,88,352	7,38,215	7,83,578	8,18,912	8,44,220
Closing balance	38,20,000	38,20,000	36,29,000	33,42,500	29,60,500	24,83,000	20,05,500	14,32,500	7,64,000	0
Cash flow available (EBITDA – tax)	-4,28,019	-1,33,259	4,13,360	12,36,790	22,85,709	27,70,416	29,12,577	31,18,439	32,54,322	33,93,802
DSCR (x)	-1.07	-0.33	0.70	1.85	3.12	3.51	3.95	3.98	3.97	4.02
Net cash after debt service	-8,29,119	-5,34,359	-1,78,740	5,69,245	15,52,747	19,82,063	21,74,362	23,34,861	24,35,410	25,49,582
Cumulative cash	-8,29,119	-13,63,477	-15,42,217	-9,72,972	5,79,774	25,61,838	47,36,200	70,71,062	95,06,471	1,20,56,053

Years 1–3. DSCR is below 1.0 (-1.07, -0.33, 0.70) and cumulative cash turns positive only in year 5. The shortfall peaks at ₹15.42 lakh, which the ₹16 lakh promoter reserve covers. This is a ramp-up risk the lender should price in.

From year 4. DSCR rises from 1.85 to 4.02; average DSCR over the repayment years is 3.1x. Typical bank threshold is 1.25–1.5x.



Return metric (base case, before financing)	Value
Project IRR, 10 years, no residual value	20.1%
NPV at 12% discount rate	₹32,20,827
Simple payback on project cost before GST	5.5 years
Break-even throughput	6.6 sessions / day (159 kWh/day)

Project cash flow = EBITDA – tax, with the investment at ₹49,14,798 (GST treated as recoverable). Equipment is assumed to have no residual value after year 10, which is conservative.

12. Sensitivity analysis

Scenario	Assumptions	Year-5 revenue (₹)	Project IRR	Min DSCR, Y3–Y10	Cumulative cash positive
Conservative	8,000 vehicles/day; EV share ×0.7; stop rate ×0.8	19,03,322	-5%	-0.58	beyond year 10
Base	12,000 vehicles/day	50,98,183	20%	0.70	Year 5
Optimistic	16,000 vehicles/day; EV share ×1.3; stop rate ×1.2	1,06,04,221	40%	2.90	Year 3

One variable at a time (base case otherwise)

Change	Year-5 revenue (₹)	Project IRR	Min DSCR, Y4–Y10
Traffic 20% lower (9,600 vehicles/day)	40,78,547	15%	1.23
Traffic 20% higher (14,400 vehicles/day)	61,17,820	25%	2.47
Selling price ₹18 per kWh	45,88,365	16%	1.37
Selling price ₹22 per kWh	56,08,002	24%	2.34
Power cost ₹8 per kWh	50,98,183	18%	1.60
Power cost ₹6 per kWh	50,98,183	22%	2.10
Running costs +25%	50,98,183	18%	1.54
EV share 30% lower	35,68,728	12%	0.92

The result swings most with traffic and EV share. A ₹1 per kWh change in selling price or power cost matters less than a 20% change in traffic.

Ways to strengthen the case. Phase the 120 kW unit after the 30 kW unit proves demand; secure tie-ups with taxi, school-bus and intercity operators; add solar on the canopy; add a café or shop; apply for PM E-DRIVE support if eligible.

13. Risks and mitigation

Risk	Rating	Effect	Mitigation
Traffic below assumption	High	Slower ramp-up, DSCR below 1.0 for longer	7-day classified count before sanction; phase the 120 kW unit; promoter reserve
Slow EV adoption in the district	High	Fewer sessions in years 1–3	Fleet and taxi tie-ups; listing on charging apps; highway tourists from outside the district
Power tariff or demand charge higher than ₹7/kWh	Medium	Margin falls ₹1 per kWh per ₹1 rise	Get written EV tariff from distribution company; solar on canopy
Competition from new fast chargers on NH-67	Medium	Lower stop rate	First-mover position; roadside visibility; reliability and uptime
Statutory delay (sanction, CEIG, road access)	Medium	Late opening, extra interest	Apply early; carry allowance for fees; work-order only after sanction
Equipment failure or downtime	Medium	Lost sessions	Exicom warranty and AMC; 6-month warranty on other equipment per quotation
Technology change (higher-power or new connectors)	Low	Obsolescence	Modular chargers; 10-year depreciation, no residual value counted
Cost overrun	Low–Medium	Higher project cost	Quoted rates; 'actuals' items held in allowance; quantity variation clause in quotation
Fire or electrical hazard	Low	Loss and liability	Fire kit, earthing, CEIG inspection, fire and third-party insurance

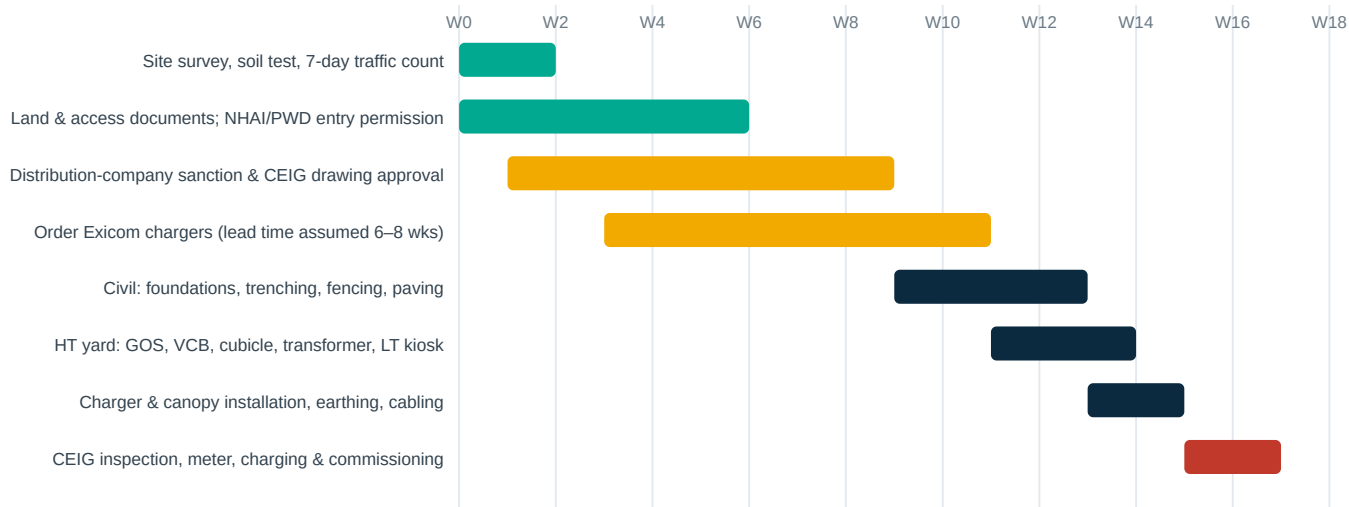
Suggested loan covenants

- Primary security: hypothecation of chargers, transformer and other plant; equitable mortgage on the plot (if owned by the borrower).
- Debt-service reserve equal to six months' interest, funded from the promoter reserve.
- Escrow of all charging revenue; insurance assigned to the bank.
- Release of the second tranche only after the traffic count, the distribution-company sanction and CEIG approval are on file.

14. Implementation plan and approvals

The quotation allows 30–45 days from receipt of the advance and site/document approvals **QUOTED**. The schedule below adds the approval and equipment lead times, which the quotation does not cover.

Indicative implementation schedule (weeks from work order and advance)



Approval	Authority	Status
Distribution-company sanction and feasibility (11 kV, 200 kVA)	Distribution company	To apply
Electrical drawing and inspection approval	CEIG, Karnataka	To apply
CT/PT meter testing; TAQC/MRT	Distribution company	To apply
Road-access / entry permission on NH-67	NHAI or PWD	To apply
Land-use / conversion status; local authority NOC	Revenue / local body	To confirm
Udyam registration; GST registration	Borrower	To confirm
Fire safety clearance (if required)	Fire department	To confirm

Operation and management

- Operated 24 hours with two attendants in shifts; payment and monitoring through a charge-point management system with OCPP-compatible chargers (to be confirmed in the Exicom quote).
- Maintenance: annual maintenance contract for chargers; periodic transformer and earthing tests.
- Warranty (per quotation): transformer 12 months; other electrical equipment 6 months from commissioning **QUOTED**.

Appendix A. Bill of quantities (The Lynx quotation, 06-10-2026)

#	Head	Description	Unit	Qty	Rate	Amount (₹)
1	Statutory	BESCOM Approval, Sanction & Clearance Charges	LOT	1	225000	2,25,000
2	Statutory	CEIG, TAQC & MRT Statutory Charges	LOT	1	80000	80,000
3	Statutory	Security deposit, CEIG drawing, inspection fees, CT/PT & HT meter testing, TAQC/MRT and other fees against receipts	EA	1	0	Actuals
4	Safety	11 kV Danger Warning Board	NOS	2	250	500
5	Safety	9 Ltr Fire Bucket with Stand, 3-bucket set	SET	1	4000	4,000
6	Safety	9 kg ABC Type Fire Extinguisher	NOS	2	4500	9,000
7	Safety	11 kV Electrical Shock Treatment Chart	NOS	1	500	500
8	Safety	HT/LT Cable Route Marker, cast iron	NOS	2	400	800
9	PVC Pipe	100 mm PVC/DWC cable protection pipe, supply and laying	MTR	40	250	10,000
10	GOS	Spun pole / DP arrangement with 400 A GOS, lightning arrester, earthing, concreting	LOT	1	144000	1,44,000
11	HT Cubicle	HT metering cubicle, supply, installation, testing, commissioning	SET	1	252000	2,52,000
12	VCB	11 kV, 630 A, 25 kA, 3P VCB panel	SET	1	351000	3,51,000
13	TC	200 kVA, 11/0.433 kV outdoor oil-filled ONAN transformer, off-load tap changer, Level-2	NOS	1	425000	4,25,000
14	Cable	3C×95 sq mm, 11 kV grade HT cable	MTR	44	1380	60,720
15	Potted	3C×95 sq mm HT cable indoor termination kit	NOS	6	5400	32,400
16	Potted	3C×95 sq mm HT cable outdoor termination kit	NOS	2	6000	12,000
17	Cable	3.5C×120 sq mm Al XLPE armoured cable	MTR	14	995	13,930
18	Termination	End termination for 3.5C×120 sq mm Al armoured cable, incl. glanding and lugs	NOS	4	3335	13,340
19	LT Kiosk Panel	LT kiosk panel 400 A, 415 V, 3P+N; 250 A 4P MCCB (120 kW) and 63 A 4P MCCB (30 kW) outgoing	SET	1	225000	2,25,000
20	Earthing	Copper-bonded pipe earthing electrode with accessories	NOS	12	12250	1,47,000
21	Earthing	50×6 mm GI earthing flat	MTR	41	400	16,400
22	Earthing	600×600×3 mm copper plate earthing for transformer neutral	NOS	2	31500	63,000
23	Earthing	25×3 mm copper earthing strip	MTR	12	1524	18,288
24	Trenching	Cable trench excavation in hard soil, with backfilling and compaction	CUM	26	2200	57,323
25	Fencing	Chain-link fencing	SQM	55	1615	88,303
26	40 mm Jelly	40 mm jelly in switchyard	LOT	1	16500	16,500
27	DAS	DAS connectivity modem	NOS	1	12000	12,000
28	Civil Work	RCC foundations for HT equipment, transformer, HT cubicle, GOS, VCB, LT kiosk	NOS	1	95000	95,000
29	Logistics	Transportation, loading and unloading of materials to site	LOT	1	50000	50,000
30	Service	PMC charges: project management, site supervision and coordination	LOT	1	100000	1,00,000
31	Cable	3.5C×185 sq mm Al armoured cable	MTR	30	1374	41,220
32	Termination	End termination for 3.5C×185 sq mm Al cable	NOS	4	2500	10,000
33	Civil Work	RCC M20 charger foundation 4700×1000×400 mm with 8 mm TMT reinforcement	NOS	1	47500	47,500
34	Civil Work	PCC foundation 1:1.5:3, 1000×1000×500 mm	NOS	1	7500	7,500
35	Cable	4C×70 sq mm copper flexible cable	MTR	10	5880	58,800

36	Termination	End termination for 4C×70 sq mm Cu cable	NOS	2	1755	3,510
37	Cable	1C×50 sq mm green copper flexible cable	MTR	12	984	11,808
38	Termination	End termination for 1C×50 sq mm Cu cable	NOS	2	565	1,130
39	Cable	4C×16 sq mm copper flexible cable	MTR	10	1212	12,120
40	Termination	End termination for 4C×16 sq mm Cu cable	NOS	2	675	1,350
41	Cable	1C×16 sq mm green copper flexible cable	MTR	12	288	3,456
42	Termination	End termination for 1C×16 sq mm Cu flexible cable	NOS	2	200	400
43	HDPE Pipe	110 mm OD HDPE cable protection pipe	MTR	70	450	31,500
44	Installation	Installation and commissioning of 120 kW dual-gun DC charger	NOS	1	6000	6,000
45	Installation	Installation and commissioning of 30 kW dual-gun DC charger	NOS	1	4500	4,500
46	Earthing	GI earth strip 50×6 mm with joint welding, nuts and bolts	MTR	20	315	6,300
47	Earthing	20 mm dia × 3 m copper-bonded chemical earthing electrode with pit cover	NOS	3	12250	36,750
48	Installation	300 kW / 630 A electrical panel installation	NOS	1	2500	2,500
49	Safety	Safety bollards with fasteners	NOS	4	350	1,400
50	Safety	Vehicle car stopper	NOS	4	950	3,800
51	Safety	Wall/ground-mounted fire extinguisher	NOS	1	5000	5,000
52	Safety	Reserved parking sign board	NOS	1	250	250
53	Debris	Site debris removal and cleaning	LOT	2	2500	5,000
54	Service	One-way transportation from vendor base to site (251–450 km)	NOS	1	10000	10,000
55	Trenching	Soft trenching in soil with backfilling	CUM	33	1600	52,800
56	Termination	PU foam for cable/conduit openings	LOT	1	700	700
57	Interlocking Tiles	Interlocking pavers over 2-inch aggregate/mortar base	SQFT	600	250	1,50,000
58	Painting	Parking/foundation floor paint, 1 coat primer + 2 finish coats, stencil marking	SQFT	600	140	84,000
59	Granite	19 mm thick granite	SQFT	150	550	82,500
Total (59 items, before GST)						32,04,798

Appendix B. Traffic-survey template (7 days, classified count at the site)

Count both directions at the plot frontage for 7 consecutive days, including one market day and one weekend. Record each hour in the slots below; record EV cars by their green number plates.

Time slot	Cars	Taxis/vans	Buses	Trucks	2W	3W	EV cars (green plate)
06:00–09:00							
09:00–12:00							
12:00–15:00							
15:00–18:00							
18:00–21:00							
21:00–06:00							

Enter the 7-day average daily total in the model's first input (12,000 vehicles per day is the current assumption).

Appendix C. Sources

India EV share 2025 (Vahan)	https://newsonair.gov.in/indias-electric-vehicle-market-accounts-for-8-of-new-vehicle-registrations-in-2025-vahan-portal-data/
Karnataka e-car sales FY 2025-26	https://evreporter.com/state-wise-electric-car-penetration-in-india-in-fy-2025-26/
Karnataka e-4W FY24, FY25; EV share	https://www.deccanherald.com/india/karnataka/bengaluru/bengaluru-all-charged-up-for-evs-but-lax-charging-infra-gives-buyers-a-shock-3684174
Karnataka road-tax change, Apr 2026	https://vahanbazaar.in/news/karnataka-ev-tax-exemption-ends-new-slabs-april-2026
PM E-DRIVE charging guidelines	https://www.outlookbusiness.com/ampstories/planet/industry/centre-launches-guidelines-for-72300-ev-charging-stations-under-pm-e-drive-scheme-details-here
KERC tariff and EV category	https://www.energetica-india.net/news/kerc-plans-cross-subsidy-and-surcharge-cuts-targets-ev-and-lift-irrigation-tariffs-among-others
Gadag district population	https://www.citypopulation.de/en/india/karnataka/admin/561__gadag
Gadag-Betageri	https://en.wikipedia.org/wiki/Gadag-Betageri
Charging points in Gadag	https://www.91trucks.com/electric/charging-stations/gadag
NH-67 route	https://en.wikipedia.org/wiki/National_Highway_67_(India)
NHAI Hubballi–Hosapete award	https://www.itln.in/road-transport-nhai-awards-contract-for-two-highway-development-projects-in-karnataka
Exicom Harmony DC chargers	https://exicom.in/newsroom/exicom-launches-indias-fastest-dc-charger-engineered-to-make-ev-charging-experience-effortless
Bank of India EV-charging MSME terms, 2024	https://www.pv-magazine-india.com/2024/07/26/tata-power-solar-partners-with-bank-of-india-for-low-cost-finance-to-rooftop-solar-and-ev-charging-station-customers/
Priority-sector lending proposal, Aug 2026	https://www.alcicle.com/news/india-banks-propose-priority-lending-for-ev-charging-and-e-fleets-120914
Imagery	Esri World Imagery (© Esri, Maxar, Earthstar Geographics)
Road name and class	OpenStreetMap contributors (ODbL)
Costs	The Lynx quotation dated 06-10-2026; charger and canopy prices and 5% GST rate supplied by the client